

Notes: Falato, Kadyrzhanova, Sim, and Steri (JF, 2022)
Rising Intangible Capital, Shrinking Debt Capacity, and the U.S. Corporate Savings Glut

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1 One-sentence summary

This note summarizes Falato, Kadyrzhanova, Sim, and Steri (2022), which uses U.S. Compustat data from 1970 to 2010, a firm-level intangible-capital measure, a calibrated dynamic corporate-finance model with tangible and intangible capital, and panel regressions including patent-pledgeability evidence to show that the rise of difficult-to-pledge intangible capital shrank firms' collateral-based debt capacity and explains most of the long-run increase in U.S. corporate cash holdings.

2 Big picture

The paper connects two major long-run changes in U.S. public corporations.

- On the **financial side**, firms have accumulated much more cash. Average cash-to-assets rose from roughly 9% in 1970 to above 20% by the late 2000s.
- On the **real side**, firms have shifted away from tangible assets toward intangible capital such as R&D, organizational capital, brand capital, and software or IT capital.
- The paper's central claim is that these two trends are connected: intangible capital is productive, but it cannot easily be pledged as collateral. As the production technology becomes more intangible, firms lose debt capacity and substitute toward internal liquidity.

The mechanism is simple but powerful. A firm with growth options expects future investment opportunities. If external finance is costly and only tangible assets can be pledged, then a firm with more intangible capital cannot borrow enough against its asset base when those opportunities arrive. Holding cash becomes a way to preserve financial flexibility.

3 Incremental contribution

3.1 What the paper adds

- It provides a quantitative explanation for the U.S. corporate savings glut based on the rise of intangible capital.
- It moves beyond reduced-form cash-holding regressions by building a calibratable dynamic model with two productive assets: tangible capital and intangible capital.
- It introduces a collateral distinction between the two capital types: tangible capital can back debt, while intangible capital cannot.
- It shows that this collateral mechanism can account for about three-quarters of the observed increase in average cash ratios.
- It also explains cross-sectional and aggregate features: small firms hold more cash, the spread between small and large firms rises, and aggregate cash rises despite concentration among large firms.

3.2 Relation to the literature

The paper contributes to three literatures.

1. **Corporate liquidity and cash holdings.** Earlier work emphasizes cash-flow risk, financing costs, taxes, agency conflicts, competition, and interest rates. This paper sizes up these forces in a structural model and argues that intangible capital is quantitatively dominant for the secular trend.
2. **Collateral and debt capacity.** The paper applies the classical collateral logic of Hart and Moore, Kiyotaki and Moore, Shleifer and Vishny, and Rampini and Viswanathan to the modern shift toward intangible production.
3. **Intangible capital in finance.** Prior work shows that intangible capital matters for investment-Q relations and asset pricing. This paper shows that intangible capital also changes firms' financing and liquidity policies.

4 Data and measurement

4.1 Sample

The empirical sample is a panel of U.S. public corporations from Compustat over 1970 to 2010. The authors exclude financial firms and utilities, and keep firm-year observations with positive book assets and sales. The main sample contains 176,877 firm-year observations for 18,535 unique firms.

4.2 Cash and leverage

- **Cash ratio:** cash and marketable securities divided by book assets.
- **Net leverage:** debt minus cash divided by book assets.
- The secular increase in cash is accompanied by a decline in net leverage, consistent with a balance-sheet shift toward liquid internal funds.

4.3 Firm-level intangible capital

The paper constructs intangible capital as the sum of three components:

1. **Knowledge capital:** accumulated R&D investment, capitalized using the perpetual inventory method with depreciation.
2. **Organizational capital:** accumulated SG&A investment, interpreted as investment in brand, organization, worker training, distribution systems, and firm-specific organizational capability.
3. **Information technology capital:** software and computerized information capital, constructed using industry-level BEA data and mapped to firms.

The key measurement choice is to construct intangible capital from investment flows rather than relying on accounting balance-sheet intangibles, because internally generated intangibles are usually expensed rather than booked as assets.

5 Model: core mechanism

5.1 Production technology

The firm produces with tangible capital K_T and intangible capital K_N :

$$\Pi(Z, K_T, K_N) = Z^{1-\gamma} \Phi(K_T, K_N)^\gamma - ZF_O,$$

where Z is idiosyncratic productivity and F_O is a fixed operating cost.

The capital aggregator is CES:

$$\Phi(K_T, K_N) = \left[\theta \left(\frac{K_T}{\theta} \right)^{-\rho} + (1 - \theta) \left(\frac{K_N}{1 - \theta} \right)^{-\rho} \right]^{-1/\rho}.$$

Here θ controls the long-run asset tangibility of production. A decline in θ means that the firm relies more heavily on intangible capital.

5.2 Capital adjustment costs

Investment in both tangible and intangible capital is subject to fixed adjustment costs. This generates (S, s) -type behavior: firms wait, accumulate liquidity, and then make lumpy investment adjustments when capital is sufficiently far from target.

This real friction is important because lumpy investment creates large financing needs. Cash is valuable because it allows the firm to finance these investment spikes without costly outside equity.

5.3 Financing frictions

The model has three sources of funds: internal funds, debt, and outside equity.

- Debt is risk-free but collateral constrained.
- Equity issuance is costly, with both fixed and linear issuance costs.
- Cash is the negative of net debt, so $B < 0$ denotes cash holdings and $B > 0$ denotes net debt.

The central borrowing constraint is

$$B' \leq \bar{B}(K'_T) \equiv \frac{(1 - \delta_T)K'_T}{1 + r(1 - \tau)}.$$

Only tangible capital enters the collateral constraint. Intangible capital is productive, but it does not expand borrowing capacity.

5.4 Economic intuition

The mechanism can be summarized in four steps.

1. Firms need cash because they face costly external equity and lumpy investment opportunities.
2. Debt is cheaper than equity but limited by tangible collateral.
3. A shift toward intangible capital reduces collateral per unit of productive capital.
4. Lower debt capacity makes cash more valuable as precautionary liquidity.

Thus, the rise of intangible capital increases cash not because intangible firms are mechanically unprofitable, but because they need liquidity while having less pledgeable collateral.

6 Quantitative strategy and main results

6.1 Calibration

The annual calibration uses 12 parameters. Some are taken from prior literature, while others are chosen to match empirical moments such as asset tangibility, profitability, volatility, equity issuance, tax rates, and real interest rates.

A useful benchmark is the polar case $\theta = 1$, in which firms use only tangible capital. In this case, the model is close to a standard dynamic corporate finance model with physical capital and costly external finance. The paper shows that this benchmark generates cash ratios far too low relative to the data.

6.2 Main quantitative counterfactual

The main experiment compares an early period, 1975 to 1985, with a late period, 1995 to 2005. The model is calibrated separately for each period. The key counterfactual holds asset tangibility fixed at its early-period level while allowing other parameters to change.

- In the data, average cash ratios rise by 8.3 percentage points.
- In the full model, average cash ratios also rise by 8.3 percentage points.
- If the rise in intangible capital is switched off, the model predicts only a 2.1 percentage-point increase.
- Therefore, the intangible-capital channel explains about 6.2 percentage points, roughly three-quarters of the observed increase.

6.3 Alternative mechanisms

The authors also switch off other channels.

- Holding cash-flow volatility fixed reduces the predicted increase in cash only modestly.
- Holding interest rates fixed also has only a small effect.
- Equity issuance costs are important for the level of cash, but their empirical decline works against the observed upward trend in cash.

6.4 Cross-sectional and aggregate implications

The model also matches the size distribution of cash.

- Small firms hold more cash than large firms.
- The spread between small-firm and large-firm cash ratios increases over time.
- The largest firms still hold most aggregate dollars of cash, so matching aggregate cash requires the mechanism to operate among large firms as well.
- The model explains a large share of the increase in both average and aggregate cash ratios.

7 Reduced-form evidence and collateral validation

7.1 Panel regressions

The paper estimates panel regressions of the cash ratio on intangible capital and standard determinants of cash. The coefficient on intangible capital is positive and statistically significant across specifications with industry fixed effects and firm fixed effects. The result is robust to alternative measures of intangibles, including R&D capital, organizational capital, and the Peters and Taylor intangible capital measure.

The reduced-form estimates suggest that more than 40% of the predicted rise in cash can be attributed to intangible capital. This is smaller than the structural model's three-quarter estimate but still economically large.

7.2 Alternative-channel controls

The paper also controls for several competing explanations: industry competition, multinational tax incentives, broad-based equity compensation, cost of carry, corporate labor share, and firm age. These variables matter, but the intangible-capital coefficient remains stable, suggesting that the baseline relation is not simply an omitted-variable artifact.

7.3 Patent pledgeability evidence

The paper provides additional evidence for the collateral mechanism. If intangible assets become more pledgeable, firms should need less cash. The authors use pledged patents and the Delaware ABSFA law as evidence. The estimates show that patent pledgeability is associated with lower cash ratios, supporting the debt-capacity mechanism.

8 Identification and interpretation

8.1 What the paper can identify

The structural exercise identifies whether a calibrated model with intangible collateral constraints can quantitatively reproduce the secular increase in cash. It does not provide a single quasi-experimental

estimate of the effect of intangible capital on cash. The paper’s credibility comes from a combination of model fit, counterfactual decomposition, panel evidence, and patent-pledgeability tests.

8.2 Main limitations

- Intangible capital is difficult to measure. The paper constructs a rich proxy, but some forms of brand, customer loyalty, and organizational knowledge are still imperfectly observed.
- Reduced-form panel estimates cannot fully rule out alternative mechanisms, especially unobserved shifts in firm type, business models, or market power.
- The model is intentionally parsimonious. It abstracts from richer debt contracts, leasing, international tax structures, heterogeneous collateral markets, and detailed firm entry and exit.
- The paper is best read as a quantitative mechanism paper rather than a clean causal identification paper.

8.3 Why the paper is important

The paper reframes the corporate savings glut as a consequence of the changing nature of production. Modern firms may not be hoarding cash only because of tax incentives, agency problems, or uncertainty. They may hold more cash because their most important assets are productive but hard to pledge.

9 Tables and figures

9.1 Figure 1: Intangible capital, cash holdings, and leverage

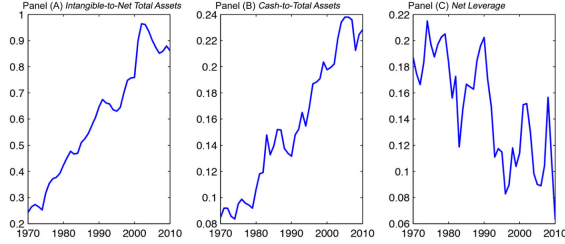


Figure 1. Intangible capital, cash holdings, and leverage. The figure depicts the long-run trend in the intangible capital ratio (relative to total assets net of cash) in Panel (A), the cash ratio (relative to total assets) in Panel (B), and the net debt ratio (relative to total assets) in Panel (C). The ratios are plotted as equal-weighted averages across all firms in the sample in each year. Total assets refer to the balance sheet book value of total assets. The sample includes all Compustat firm-year observations from 1970 to 2010 with positive values for the book value of total assets and sales revenue for firms incorporated in the United States. Financial firms (SIC code 6000-6999) and utilities (SIC codes 4900-4999) are excluded from the sample, yielding a panel of 176,877 observations for 18,535 unique firms. Variable definitions are provided in Appendix B. (Color figure can be viewed at wileyonlinelibrary.com)

Read: The figure shows three long-run trends: intangible capital rises sharply, cash holdings rise steadily, and net leverage falls. The rise in intangible capital and cash is visually synchronized over the sample period.

Economic message: The figure motivates the paper's core mechanism: as firms become more intangible, they substitute away from debt capacity and toward internal liquidity.

9.2 Figure 2: Cash holdings in subsamples by size, industry, and age

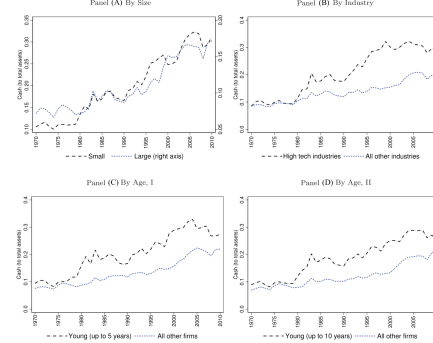


Figure 2. Cash holdings in subsamples by size, industry, and age. The figure depicts the long-run trend in the cash ratio in subsamples split by size (above and below median by total firm assets, Panel (A)), whether the firm is in a high-tech industry as defined by Loughran and Ritter (2004) (Panel (B)), and age (less than five years since IPO (Panel (C)) and less than 10 years since IPO (Panel (D)) versus all other firms). The sample includes all Compustat firm-year observations from 1970 to 2010 with positive values for the book value of total assets and sales revenue for firms incorporated in the United States. Financial firms (SIC code 6000-6999) and utilities (SIC codes 4900-4999) are excluded from the sample, yielding a panel of 176,877 observations for 18,535 unique firms. Variable definitions are provided in Appendix B. (Color figure can be viewed at wileyonlinelibrary.com)

Read: Cash ratios rise broadly across firm groups, but the trend is stronger for small firms, high-tech firms, and younger firms.

Economic message: The rise in cash is not only an aggregate phenomenon. It is strongest among firms that are more likely to be intangible-intensive and financially constrained.

9.3 Figure 3: Intangible capital in subsamples by size, industry, and age

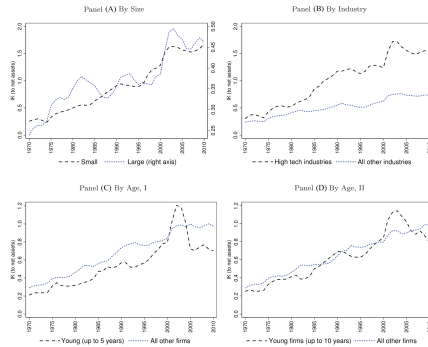


Figure 3. Intangible capital in subsamples by size, industry, and age. The figure depicts the long-run trend in the intangible capital ratio in subsamples split by size (above and below median by total firm assets, Panel (A)), whether the firm is in a high-tech industry as defined by Loughran and Ritter (2004) (Panel (B)), and age (less than five years since IPO (Panel (C)) and less than 10 years since IPO (Panel (D)) versus all other firms). The sample includes all Compustat firm-year observations from 1970 to 2010 with positive values for the book value of total assets and sales revenue for firms incorporated in the United States. Financial firms (SIC code 6000-6999) and utilities (SIC codes 4900-4999) are excluded from the sample, yielding a panel of 176,877 observations for 18,535 unique firms. Variable definitions are provided in Appendix B. (Color figure can be viewed at wileyonlinelibrary.com)

Read: Intangible capital rises across size, industry, and age groups, with especially strong increases among high-tech firms and smaller or younger firms.

Economic message: The same groups with stronger cash accumulation also experience stronger growth in intangible capital, reinforcing the proposed link.

9.4 Figure 4: Cash, aggregate facts

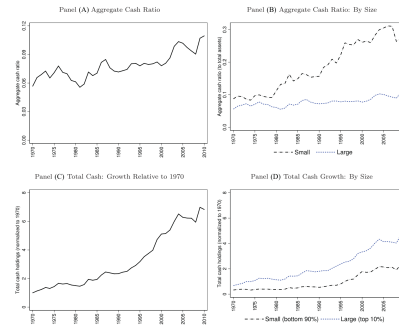


Figure 4. Cash, aggregate facts. The figure depicts the long-run trend in the aggregate cash ratio and aggregate cash holdings. Panel (A) plots the aggregate cash ratio, defined as the ratio of total cash holdings to total assets across all firms in each year. Panel (B) plots the long-run trend in the aggregate cash ratio in subsamples split by size (above and below median by total firm assets). Panel (C) plots the growth in annual aggregate cash holdings relative to 1970, defined as the sum of cash across all firms in the sample in each year normalized by the value of this sum in the beginning of sample (1970). Panel (D) plots the growth in annual aggregate cash holdings relative to 1970 for the largest decile of firms in each year versus the rest of the sample. The sample includes all Compustat firm-year observations from 1970 to 2010 with positive values for the book value of total assets and sales revenue for firms incorporated in the United States. Financial firms (SIC code 6000-6999) and utilities (SIC codes 4900-4999) are excluded from the sample, yielding a panel of 176,877 observations for 18,535 unique firms. Variable definitions are provided in Appendix B. (Color figure can be viewed at wileyonlinelibrary.com)

Read: Aggregate cash ratios and aggregate cash levels rise over time, although the level of aggregate cash is concentrated among the largest firms.

Economic message: A successful explanation must match not only the average firm but also the aggregate distribution of cash.

9.5 Figure 5: Optimal policies

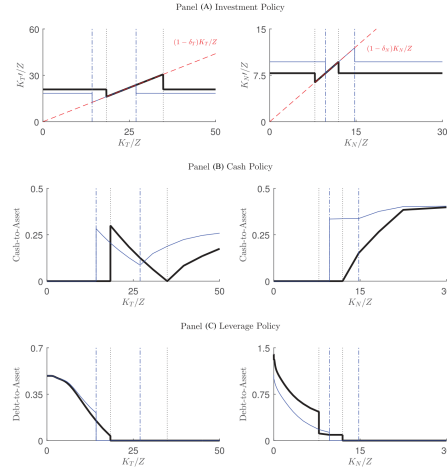


Figure 5. Optimal policies. The figure depicts the optimal investment (Panel A), liquidity (Panel B), and leverage policies (Panel C) as a function of tangible and intangible capital stocks, K_T/Z and K_S/Z . The other state variables are set to their steady-state values. The thick black lines refer to high asset tangibility ($\theta = 0.7$), while the thin blue lines refer to low asset tangibility ($\theta = 0.6$). The vertical black dotted lines delimit the inaction region for the case with high asset tangibility, while the vertical blue dashed-dotted lines delimit the inaction region for the case with low asset tangibility. All remaining parameters are set to their baseline values for the full sample as in Table 1. (Color figure can be viewed at wileyonlinelibrary.com)

Read: The policy functions show investment inaction regions, cash accumulation regions, and leverage policies under high versus low asset tangibility. Lower tangibility shifts firms toward more cash and less debt.

Economic message: This figure is the model mechanism in graphical form: intangible capital weakens collateral capacity, making cash more valuable before investment adjustments.

9.6 Table I: Baseline Calibration

Table I Baseline Calibration		
The table reports the calibrated structural parameters. The frequency of the calibration is annual. P^O is reported as a proportion of average tangible capital in the economy.		
Description	Calibration	Target/Restriction
Technological Parameters		
Curvature of profit function	$\gamma = 0.600$	Hennessey and Whited (2005), Hennessey and Whited (2007)
Asset tangibility	$\theta = 0.660$	Average mix of tangible and intangible capital (data)
Depreciation of tangible capital	$\delta_T = 0.120$	Average depreciation of intangible capital (data)
Depreciation of intangible capital	$\delta_I = 0.190$	Average depreciation of intangible capital (data)
Elasticity of substitution between capital inputs	$\rho = 1.000$	High elasticity of substitution ($=0.5$)
Fixed cost of adjustment	$P^A = 0.005$	Bloom (2009)
Fixed cost of operation	$P^O = 0.085$	Average operating profits (data)
Volatility of technology shock	$\sigma_z = 0.350$	Idiosyncratic volatility of operating profits growth (data)
Financial Parameters		
Risk-free rate	$r = 0.011$	Average real interest (three-month Treasury) rate (data)
Fixed cost of issuance	$\phi_0 = 0.011$	Bolton, Chen, and Wang (2011), Altinkilic and Hansen (2000)
Linear cost of issuance	$\phi_1 = 0.015$	Average equity issuance (data)
Tax rate	$\tau = 0.200$	Nikolov and Whited (2014)

Read: The table lists the structural parameters used in the baseline model, including the profit curvature, asset tangibility, depreciation rates, adjustment costs, fixed operating costs, volatility, interest rate, equity issuance costs, and tax rate.

Economic message: The calibration is disciplined by prior literature and empirical moments, which makes the quantitative counterfactual interpretable.

9.7 Table II: Asset Tangibility and Cash Holdings: Baseline Quantitative Results

Table II
Asset Tangibility and Cash Holdings: Baseline Quantitative Results

This table reports model-implied and data moments for the baseline calibration. The data moments in the first column are for a sample of nonfinancial, unregulated firms from the annual Compustat, which we complement with information on equity issuance from the Securities Data Company (SDC) database. The sample period is 1970 to 2010. The model-implied moments in the second column are calculated as averages of simulations of 1,000 firms and 100 time periods. To perform the simulation, we use 1,000 by 100 random draws from the lognormal distribution. The same set of random draws is applied to each economy. The expectation is approximated with 30-point Gauss-Hermite quadrature. The parameter values are as in Table I. The third column reports model-implied moments for the case in which firms use only tangible assets. All moments are self-explanatory, except for the Error Variance of Profits and Cash Autocorrelation, which are defined as the error variance and the slope coefficient from a first-order autoregression of idiosyncratic profit growth and the cash ratio, respectively. Details on model and data variable definitions are in Appendix B.

	Data (1970 to 2010)	Model (Baseline)	Model ($\theta = 1$)
Panel A: Main Moments			
Average Cash Ratio	0.140	0.151	0.020
Average Leverage Ratio	0.239	0.211	0.251
Average Equity Issuances	0.006	0.004	0.006
Frequency of Equity Issuances	0.035	0.043	0.031
Average Asset Tangibility	0.718	0.711	1.000
Average Operating Profitability	0.151	0.151	0.145
Panel B: Additional Moments			
Error Variance of Profits	0.003	0.002	0.002
Cash Autocorrelation	0.424	0.563	0.334
Skewness of Tangible Investment	1.572	2.124	1.974
Skewness of Intangible Investment	1.721	2.149	—

Read: The baseline model matches average cash, leverage, equity issuance, asset tangibility, and profitability. The tangibles-only benchmark generates cash ratios that are far too small.

Economic message: Intangible capital is not a small refinement. It is essential for matching corporate cash holdings quantitatively.

9.8 Table III: Main Quantitative Counterfactual: Calibration

Table III
Main Quantitative Counterfactual: Calibration

This table reports the calibrated structural parameters for the main quantitative counterfactual, which involves a comparison between model-implied and data moments for the 1975 to 1985 decade ("Early Period") and the 1995 to 2005 decade ("Late Period").

Description	Calibration	
	Early (1975 to 1985)	Late (1995 to 2005)
Technological Parameters		
Asset tangibility (θ)	0.710	0.620
Fixed cost of operation (F^O)	0.056	0.119
Volatility of technology shock (σ_z)	0.340	0.360
Financial Parameters		

Read: The table reports early-period and late-period parameter values. Asset tangibility falls, operating costs rise, volatility changes modestly, interest rates decline slightly, and equity issuance costs fall.

Economic message: The counterfactual framework lets the paper separate the intangible-capital channel from other changing forces.

9.9 Table IV: Asset Tangibility and the Rise in Cash Holdings: Results of Main Quantitative Counterfactual

Table IV
Asset Tangibility and the Rise in Cash Holdings: Results of Main Quantitative Counterfactual

This table reports model-implied and data moments for the main quantitative counterfactual, which involves a comparison of data and model-implied moments between the "Early Period" (1975 to 1985) and the "Late Period" (1995 to 2005). The data moments in the first and third columns correspond to a sample of nonfinancial, unregulated firms from the annual Compustat, which we complement with information on equity issuance from SDC. The model-implied moments in the second and fourth columns are calculated as averages of simulations of 1,000 firms and 100 time periods. To perform the simulation, we use 1,000 by 100 random draws from the lognormal distribution. The same set of random draws is applied to each economy. The expectation is approximated with 30-point Gauss-Hermite quadrature. The parameter values are as in Table I. The fifth column reports model-implied moments for a counterfactual exercise in which the tangibility parameter (θ) is kept constant at its calibrated value for the early period (1975 to 1985). Details on model and data variable definitions are in Appendix B.

	Data (1975 to 1985)	Model (1975 to 1985)	Data (1995 to 2005)	Model (1995 to 2005)	Counterfactual (θ Fixed)
Average Cash Ratio	0.100	0.095	0.183	0.178	0.116
Average Leverage Ratio	0.256	0.204	0.221	0.191	0.210
Average Equity Issuances	0.004	0.003	0.007	0.005	0.004
Frequency of Equity Issuances	0.029	0.027	0.041	0.064	0.055
Average Asset Tangibility	0.754	0.754	0.677	0.672	0.751
Average Operating Profits	0.168	0.164	0.133	0.133	0.133
Change in Cash—Model (Late vs Early): 0.083					
Change in Cash—Data (Late vs Early): 0.083					
Change due to Intangible Cap. (Counterfactual): 0.062					
Change in Leverage—Model (Late vs Early): -0.012					
Change in Leverage—Data (Late vs Early): -0.035					
Change due to Intangible Cap. (Counterfactual): -0.019					
Change in Equity Issuance—Model (Late vs Early): 0.002					
Change in Equity Issuance—Data (Late vs Early): 0.003					
Change due to Intangible Cap. (Counterfactual): 0.001					

Read: The full model reproduces the 8.3 percentage-point increase in average cash. Holding tangibility fixed leaves only a 2.1 percentage-point increase, implying 6.2 percentage points due to intangible capital.

Economic message: This is the headline quantitative result: rising intangible capital accounts for about three-quarters of the secular increase in average cash ratios.

9.10 Table V: Alternative Explanations for the Rise in Cash Holdings: Results of Additional Quantitative Counterfactuals

Table V
Alternative Explanations for the Rise in Cash Holdings: Results of Additional Quantitative Counterfactuals

This table reports model-implied and data moments for additional quantitative counterfactuals. We consider three alternative explanations: risk, financing costs, and interest rates. The data moments correspond to a sample of nonfinancial, unregulated firms from the annual Compustat, which we complement with information on equity issuance from SDC. The data moments for the late period are reported in the third column of Table II. The model-implied moments in each column are calculated as averages of simulations of 1,000 firms and 100 time periods. To perform the simulation, we use 1,000 by 100 random draws from the lognormal distribution. The same set of random draws is applied to each economy. The expectation is approximated with 30-point Gauss-Hermite quadrature. Each column reports model-implied moments for a counterfactual exercise in which, respectively, the profitability shock volatility (σ_π), the equity issuance costs (ψ_0 and ψ_1), and the risk-free rate (r) are kept constant at their calibrated value in the early period (1975 to 1985). All other parameter values are otherwise set as in Table I. Details on model and data variable definitions are in Appendix Table B.

	Counterfactual Volatility (σ_π Fixed)	Counterfactual Issuance Costs (ψ_0, ψ_1 Fixed)	Counterfactual Interest Rate (r Fixed)
Average Cash Ratio	0.164	0.297	0.191
Average Leverage Ratio	0.196	0.162	0.191
Average Equity Issuances	0.004	0.022	0.008
Frequency of Equity Issuances	0.057	0.035	0.065
Average Asset Tangibility	0.669	0.671	0.674
Average Operating Profits	0.132	0.133	0.133
Change in Cash—Data (Late vs Early):	0.070	0.203	0.096
Change due to Alternatives (Counterfactual):	0.014	-0.119	-0.013
Change in Leverage—Data (Late vs Early):	-0.007	-0.041	-0.012
Change due to Alternatives (Counterfactual):	-0.005	0.029	-0.000
Change in Equity Issuance—Data (Late vs Early):	0.002	0.019	0.006
Change due to Alternatives (Counterfactual):	0.000	-0.017	-0.004

Read: The table switches off volatility, equity issuance costs, and interest rate changes. Volatility explains a relatively small portion of the cash increase, and lower issuance costs actually work against the upward trend. **Economic message:** Alternative forces matter for levels, but they do not dominate the secular trend. Intangibility remains the main driver.

9.11 Table VI: Asset Tangibility and Cash Holdings: Quantitative Analysis of the Cross-sectional Implications

Table VI
Asset Tangibility and Cash Holdings: Quantitative Analysis of the Cross-sectional Implications

This table reports model-implied and data moments for the baseline quantitative analysis of the cross-sectional implications, which are calculated by sorting firms based on deciles of the firm size distribution. The data moments in the first and third columns correspond to a sample of nonfinancial, unregulated firms from the annual Compustat between 1970 and 2010, which we complement with information on equity issuance from SDC. The model-implied moments in the second and fourth columns are calculated as averages of simulations of 1,000 firms and 100 time periods. To perform the simulation, we use 1,000 by 100 random draws from the lognormal distribution. The same set of random draws is applied to each economy. The expectation is approximated with 30-point Gauss-Hermite quadrature. The parameter values are as in Table I. The fifth column reports model-implied moments for the value of cash, which are calculated similar to Bolton, Chen, and Wang (2011) and Gamba and Triantis (2008) as the percentage gain in firm value from having the option to hold cash, that is, as the percentage difference between firm value in the baseline and constrained value when not allowed to hold cash. Details on model and data variable definitions are in Appendix Table B.

Size	Average Cash Ratio		Aggreg. Cash Shares		Value of Cash
Decile	Data (1970 to 2010)	Model Baseline	Data (1970 to 2010)	Model Baseline	Model Baseline
Small	0.154	0.173	0.002	0.000	14.868
2	0.147	0.165	0.004	0.000	14.239
3	0.144	0.162	0.007	0.001	14.006
4	0.134	0.159	0.011	0.003	13.791
5	0.132	0.157	0.016	0.007	13.743
6	0.124	0.157	0.026	0.017	13.618
7	0.117	0.156	0.039	0.035	13.549
8	0.105	0.152	0.062	0.071	13.176
9	0.094	0.131	0.127	0.127	11.300
Large	0.080	0.102	0.707	0.738	9.590
Cross-Sectional Cash Spread (Small vs. Large)—Data: 0.074					
Cross-Sectional Cash Spread (Small vs. Large)—Model: 0.071					

Read: The model matches the higher cash ratios of small firms and the strong concentration of aggregate cash among large firms.

Economic message: The model fits both the average and the distribution, which strengthens the quantitative credibility of the mechanism.

9.12 Table VII: Asset Tangibility and the Increase in Cash Holdings: Quantitative Analysis of the Cross-Sectional and Aggregate Implications

Table VII
Asset Tangibility and the Increase in Cash Holdings: Quantitative Analysis of the Cross-Sectional and Aggregate Implications

This table reports model-implied and data moments for the quantitative analysis of the increase in cash holding for firms in each decile of the firm size distribution. The data moments in the first and third columns correspond to a sample of nonfinancial, unregulated firms from the annual Compustat, which we complement with information on equity issuance from SDC. The model-implied moments in the second and fourth columns are calculated as averages of simulations of 1,000 firms and 100 time periods. To perform the simulation, we use 1,000 by 100 random draws from the lognormal distribution. The same set of random draws is applied to each economy. The expectations is approximated with 30-point Gauss-Hermite quadrature. The parameter values are as in Table I. Details on model and data variable definitions are in Appendix Table B.

Size Decile	Average Cash Ratio				Aggreg. Cash Shares			
	Data (1975 to 1985)	Model (1975 to 1985)	Data (1995 to 2005)	Model (1995 to 2005)	Data (1975 to 1985)	Model (1975 to 1985)	Data (1995 to 2005)	Model (1995 to 2005)
Small	0.116	0.106	0.181	0.201	0.002	0.000	0.001	0.000
2	0.108	0.103	0.188	0.191	0.004	0.001	0.003	0.000
3	0.103	0.099	0.199	0.183	0.006	0.002	0.007	0.001
4	0.092	0.100	0.199	0.189	0.009	0.005	0.013	0.003
5	0.091	0.099	0.183	0.186	0.013	0.011	0.021	0.008
6	0.088	0.097	0.168	0.184	0.020	0.023	0.031	0.018
7	0.082	0.099	0.150	0.184	0.032	0.040	0.047	0.040
8	0.087	0.093	0.118	0.177	0.056	0.091	0.067	0.083
9	0.075	0.082	0.105	0.156	0.112	0.164	0.139	0.155
Large	0.069	0.066	0.086	0.127	0.146	0.656	0.670	0.691

Change in Cash (Data - Median Firm): 0.092
Change in Cash (Model - Median Firm): 0.087
Change in Cross-Sectional Cash Spread (Data - Small vs. Large): 0.048
Change in Cross-Sectional Cash Spread (Model - Small vs. Large): 0.034
Change in Aggregate Cash Ratio (Data): 0.042
Change in Aggregate Cash Ratio Due Intangible Capital & Top 5% Largest Firms (Counterfactual): 0.033
Change in Aggregate Cash Ratio Due Intangible Capital & Top 10% Largest Firms (Counterfactual): 0.039

Read: The table compares early and late periods across size deciles. The model matches the increase in median-firm cash, the widening small-large cash spread, and much of the aggregate cash increase.

Economic message: The mechanism is not limited to small firms. It is strong enough to explain a meaningful portion of aggregate cash growth.

9.13 Table VIII: Panel Evidence on Intangible Capital and Firm Financing

Table VIII
Panel Evidence on Intangible Capital and Firm Financing

This table reports estimates from panel regressions of the cash ratio on different measures of intangible capital for panel specifications with industry and firm fixed effects. In Panel A, intangible capital is the baseline IK measure defined as the sum of the stocks of past investments in firms' organizational capabilities and brand equity (organizational capital), technological knowledge (knowledge capital), and software and database development (IT capital) (see Section I for details). Panel B reports results from using only technological knowledge to measure intangibles. Panel C reports results from using only organizational capital to measure intangibles. Panel D reports results from using the Peters and Taylor (2017) measure of intangible capital. Reported coefficients in all panels are the change in the dependent variable associated with a one-standard-deviation change in the corresponding measure of intangible capital. Columns (1) and (2) correspond to the entire sample, while columns (3) and (4) correspond to the subsample of firms with positive R&D. Year dummies as well as firm-level controls for standard determinants of financial policies are included in all regressions (see Table IA.III for the full set of coefficient estimates on firm-level controls with the baseline measure). Firm fixed-effect regressions exclude firms with less than five years of data. ρ Values are in parentheses and are clustered at the firm level. Predicted change in cash due to change in intangible capital is obtained by taking the point estimates from the regression estimated over the 1970 to 1989 period and multiplying them by the difference in average value of intangible capital between the estimation (1970 to 1989) and the postestimation 1990 to 2010 period. Detailed variable definitions are in Appendix B.

	Full Sample		R&D=0 Firms	
	(1)	(2)	(3)	(4)
Baseline IK measure				
Intangible Capital	0.084*** (0.002)	0.083*** (0.003)	0.099*** (0.002)	0.098*** (0.003)
% Pred rise	42.50%		43.30%	
Pred rise	0.069		0.075	
Observations	150,574	96,136	77,052	53,858
Adjusted R^2	0.353	0.732	0.367	0.754
Only R&D Capital				
Intangible Capital	0.083*** (0.002)	0.066*** (0.003)	0.103*** (0.002)	0.087*** (0.003)
Observations	150,574	96,136	77,052	53,858
Adjusted R^2	0.336	0.716	0.388	0.742
Only Organizational Capital				
Intangible Capital	0.052*** (0.002)	0.077*** (0.002)	0.056*** (0.002)	0.090*** (0.003)
Observations	150,574	96,136	77,052	53,858
Adjusted R^2	0.265	0.732	0.296	0.753
Peters and Taylor (2017) measure				
Intangible Capital	0.087*** (0.002)	0.077*** (0.003)	0.107*** (0.002)	0.091*** (0.003)
Observations	150,574	96,136	77,052	53,858
Adjusted R^2	0.339	0.727	0.372	0.748

Read: Panel regressions show a positive relation between cash and several measures of intangible capital, with and without firm fixed effects. The result holds for the full sample and for R&D-active firms.

Economic message: The reduced-form evidence supports the model's key state variable: firms with more intangible capital hold more cash.

9.14 Table IX: Panel Evidence on Alternative Channels, Part 1

Table IX
Panel Evidence on Alternative Channels

The table reports parameter estimates from panel regressions of the cash ratio on intangible capital and a set of determinants (X) that have been previously shown to impact the cash ratio. Columns (1) and (2) show results from controlling for industry competition as proxied by the Herfindahl-Hirschman index of firm sales within three-digit Standard Industrial Classification (SIC) industries. Columns (3) and (4) show results from controlling for the implied repatriation tax penalty as constructed in Foley et al. (2007). Columns (5) and (6) show results from controlling for deferred employee pay, measured as the ratio of broad-based option compensation plans to total assets. Columns (7) and (8) show results from controlling for the cost of carry as in Azar, Kagey, and Schmalz (2016). Columns (9) and (10) show results from controlling for higher profit share of the corporate sector in the aggregate economy, as proxied by the inverse of the corporate labor share. Columns (11) and (12) show results from adding a control for firm age. Reported coefficients are the change in the dependent variable associated with a one-standard-deviation change in the explanatory variable. Other firm-level controls are as in Table IA.III. In addition, regressions in columns (3) and (4) control for the foreign income ratio as in Foley et al. (2007). Regressions in columns (1) to (6), (11), and (12) include year fixed effects, while those in columns (7) to (10) include controls for the time trend up to third order, as in Azar, Kagey, and Schmalz (2016). Firm fixed-effect regressions exclude firms with less than five years of data. *p*-Values are in parentheses and are clustered at the firm level. The predicted change in cash due to change in each determinant *X* is obtained by taking the point estimates from the regression estimated over the 1970 to 1989 period and multiplying them by the difference in average value of the variable between the estimation (1970 to 1989) and the postestimation (1989 to 2010) period. Variable definitions are in Appendix B.

X Controls for											
Industry Competition		Multinationals		Broad-Based Equity Comp		Cost of Carry		Decline in Labor Share		Firm Age	
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
Intangible Capital	0.088*** (0.002)	0.089*** (0.003)	0.094*** (0.004)	0.096*** (0.003)	0.083*** (0.004)	0.088*** (0.002)	0.089*** (0.003)	0.088*** (0.002)	0.089*** (0.003)	0.092*** (0.002)	0.089*** (0.003)

(Continued)

Read: The first part of Table IX adds controls for alternative channels such as competition, multinational taxes, broad-based equity compensation, cost of carry, labor share, and firm age.

Economic message: The intangible-capital coefficient remains positive and stable even after adding competing explanations.

9.15 Table IX: Panel Evidence on Alternative Channels, Part 2

Table IX—Continued

X Controls for							
Broad-Based Equity Comp		Cost of Carry		Decline in Labor Share		Firm Age	
(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
−0.001 (0.001)	−0.002** (0.001)	−0.003*** (0.001)	−0.006*** (0.001)	−0.006*** (0.001)	−0.005*** (0.001)	−0.027*** (0.001)	−0.009*** (0.003)
−0.7%		11.2%		2.1%		−1.4%	
72,514 0.226	47,076 0.700	150,574 0.331	95,741 0.731	144,108 0.329	92,306 0.733	131,102 0.364	95,741 0.733
Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Yes	No	Yes	No	Yes	No	Yes	No
No	Yes	No	Yes	No	Yes	No	Yes

Read: The continuation reports the additional controls, predicted contributions, observations, fixed effects, and adjusted R^2 .

Economic message: Many alternative channels matter, but they do not absorb the intangible-capital effect.

9.16 Table X: Additional Panel Evidence on Patent Pledgeability

Table X
Additional Panel Evidence on Patent Pledgeability

The table reports parameter estimates from panel regressions of the cash ratio on intangible capital and controls for the ability of a firm to pledge patents as collateral. Panel A shows results from including a firm's number of pledged patents as control. Panel B shows results from a difference-in-differences specification around the 2002 passage of the ABSFA law in Delaware as in Mann (2018), where we allow the effect of the law to vary by firms' intangible capital ratio. Reported coefficients are the change in the cash ratio associated with a one-standard-deviation change in the explanatory variable. Firm fixed-effect regressions in Panel A exclude firms with less than five years of data. The sample in Panel A spans the entire sample period (1970 to 2010), while that in Panel B is limited to eight years before and after the year of ABSFA passage. Columns (1) and (2) correspond to the entire sample, while columns (3) and (4) correspond to the subsample of firms with positive R&D. Year dummies as well as firm-level controls for standard determinants of financial policies are included in all regressions and are omitted for brevity. *p*-Values are in parentheses and are clustered at the firm level. Variable definitions are in Appendix B.

	Full Sample		R&D Firms Only	
	(1)	(2)	(3)	(4)
Panel A: Controlling for Pledged Patents				
Intangible Capital	0.089*** (0.002)	0.089*** (0.003)	0.092*** (0.002)	0.094*** (0.003)
Pledged patents	-0.061*** (0.004)	-0.016*** (0.003)	-0.079*** (0.005)	-0.022*** (0.004)
Observations	150,574	95,741	77,052	53,645
Adjusted R^2	0.334	0.733	0.369	0.754
Firm Controls	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Industry FE	Yes	No	Yes	No
Firm FE	No	Yes	No	Yes
Panel B: Difference-in-Differences Analysis				
Intangible Capital	0.085*** (0.003)	0.087*** (0.003)	0.093*** (0.003)	0.094*** (0.003)
Treated	-0.007 (0.005)	-0.001 (0.005)	-0.017*** (0.006)	-0.013* (0.007)
Intangible Capital*Treated		-0.006** (0.003)		-0.003 (0.003)
Observations	62,217	62,217	35,778	35,778

Read: Firms with more pledged patents hold less cash, and Delaware's ABSFA treatment is associated with lower cash, especially for patent-intensive firms.

Economic message: This table provides collateral-based validation: when some intangible assets become more pledgeable, the need to hold cash falls.

10 Short summary

- U.S. firms have become more intangible and have accumulated more cash over the same period.
- The paper argues that this is not a coincidence: intangible capital is productive but hard to pledge, so it shrinks debt capacity.
- In a dynamic model with lumpy investment and costly outside finance, lower pledgeability raises the value of precautionary cash.
- The calibrated model explains roughly three-quarters of the increase in average cash ratios and a large fraction of cross-sectional and aggregate cash trends.
- Panel regressions and patent pledgeability evidence support the collateral mechanism.

11 Conclusion

11.1 Core conclusion

The paper's main conclusion is that the rise of intangible capital is a quantitatively important explanation for the U.S. corporate savings glut. The shift toward intangible assets reduces firms' ability to borrow against their productive capital, and firms respond by holding more cash.

11.2 Economic intuition

The modern firm may be rich in productive assets but poor in pledgeable assets. Because intangible assets are difficult to verify, liquidate, or transfer, they are weak collateral. As firms rely more on these assets, debt capacity falls relative to investment needs. Cash becomes the substitute collateral-like buffer that preserves financial flexibility.

11.3 Takeaway

The paper changes how one should interpret high cash balances. Corporate cash is not only a passive residual or tax-driven stockpile. It can be an endogenous response to the changing production technology of firms.